

Low-cost sensors

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VITO

Introduction

Why sensors?

- Sensors are potentially interesting complementary measuring devices to fixed AQMS
- Sensors are cheap compared to fixed AQMS with reference-grade instruments
- Sensors have a high temporal resolution and can be deployed in denser networks
- Reliability of sensor data?



Sensors versus passive samplers

- On-line data
 - High time resolution ($< 1\text{h}$) -> time trends
 - Real-time visualization of results
 - Quality is discussed, no standard to evaluate sensors
 - Easy to install? (= function of system)
 - Follow-up quality of sensors
- Off-line analysis
 - Low time resolution (weeks, months)
 - No real-time visualisation
 - Accepted quality, standardized method (fct of pollutant)
 - Easy to deploy
 - No real maintenance needs

Sensors versus passive samplers

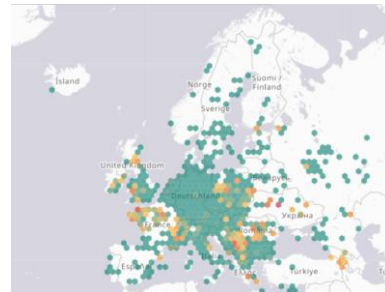


Sensor systems

What is included? -> sensors - sensor systems - sensor solutions - hybrid systems - ...



Sensors
Different types
Measuring unit

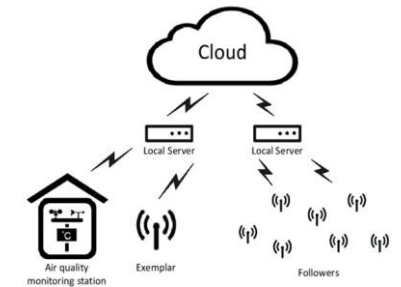


DIY building blocks
sensor, Arduino,
dataportal



Sensor systems
Sensor solutions

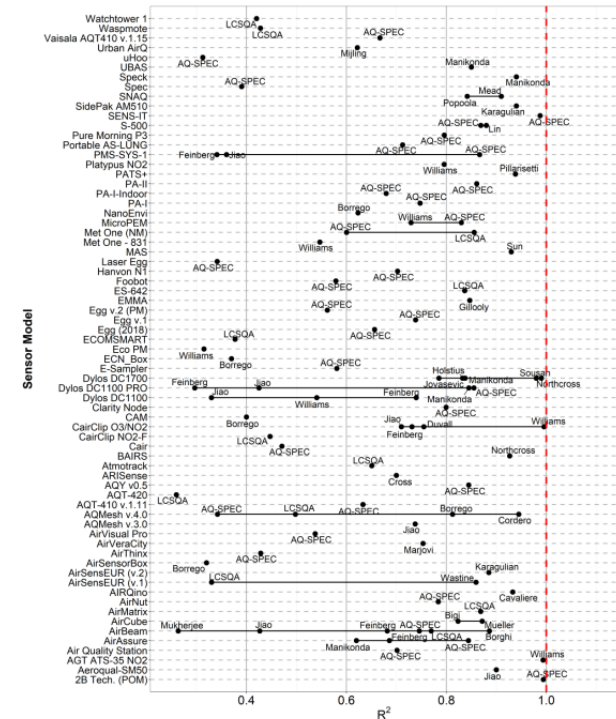
Sensor unit, database and dataportal,
calibration factors
visualisation



Hybrid systems
Sensor network
using also additional data
(a.o. AQMN, models)

How reliable are sensor systems?

- A lot of information is available:
 - Initiatives to test sensors
 - Literature on performance of sensors
- Comparison is difficult due to differences in:
 - Evaluation parameters
 - Test set-up
 - Test conditions
- Data processing
 - What data processing is used?



Karagulian et al., Atmosphere 2019, 10, 506

What do we need?

Data producers and users must understand their objectives for using LCS data in terms of how these data are expected to contribute to air quality management decision-making. Products incorporating LCS data can then be evaluated according to those objectives, ensuring that the results are relevant and useful for the intended needs. Data quality should be conducted using well defined, commonly accepted, and repeatable methods and metrics. The results of these evaluations, as well as pertinent details of the methodology used to create the data, should be disclosed in accessible, transparent ways.

- ➔ There are different applications for AQ sensors
- ➔ Different application require different data quality -> Fit for purpose!
- ➔ Also the methodology and set-up of sensor studies is important

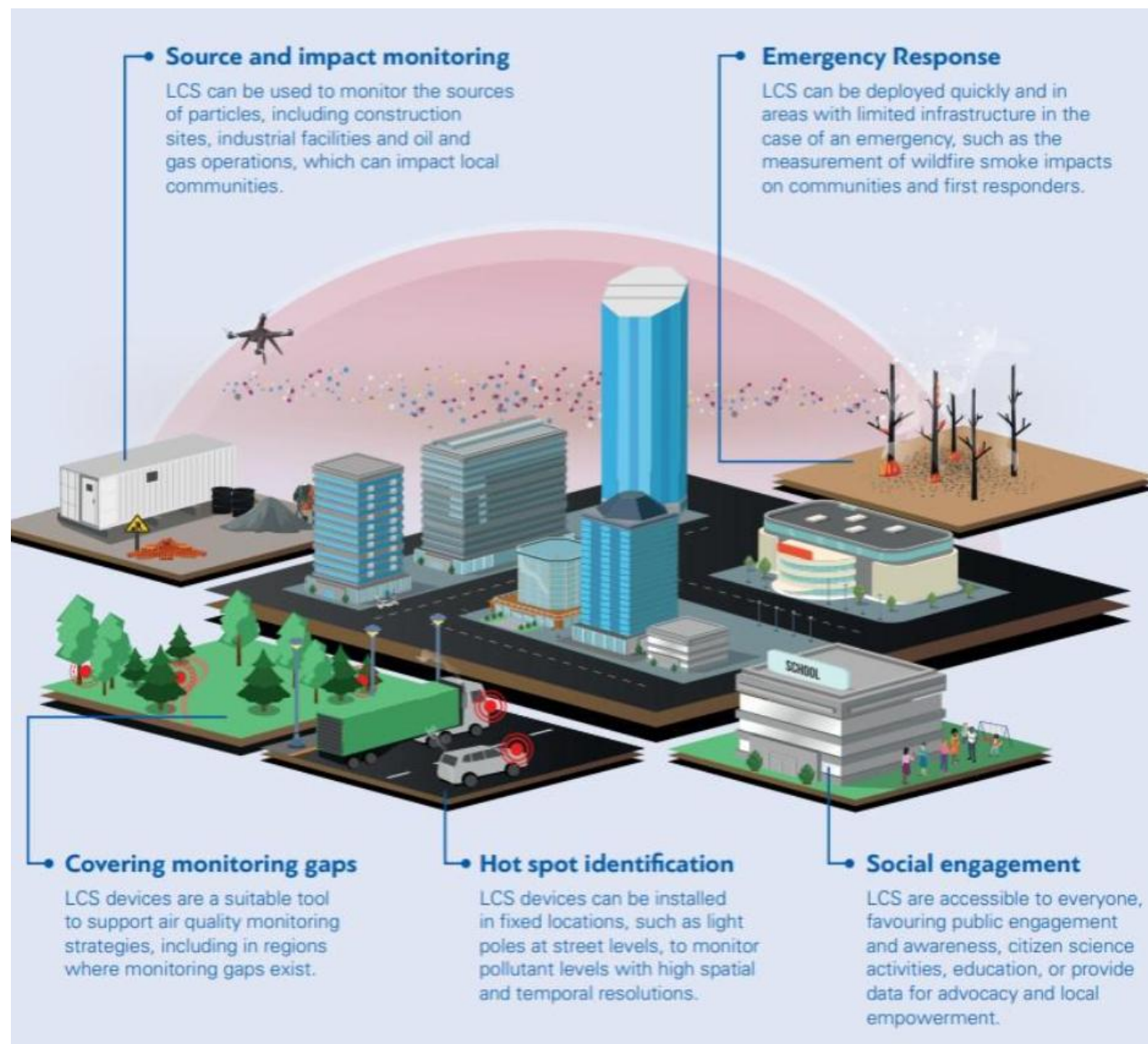


What do we need? Use cases

- Requirements of sensor quality depend on the use case
- Example for accuracy and precision



Some examples of use cases for AQ sensors

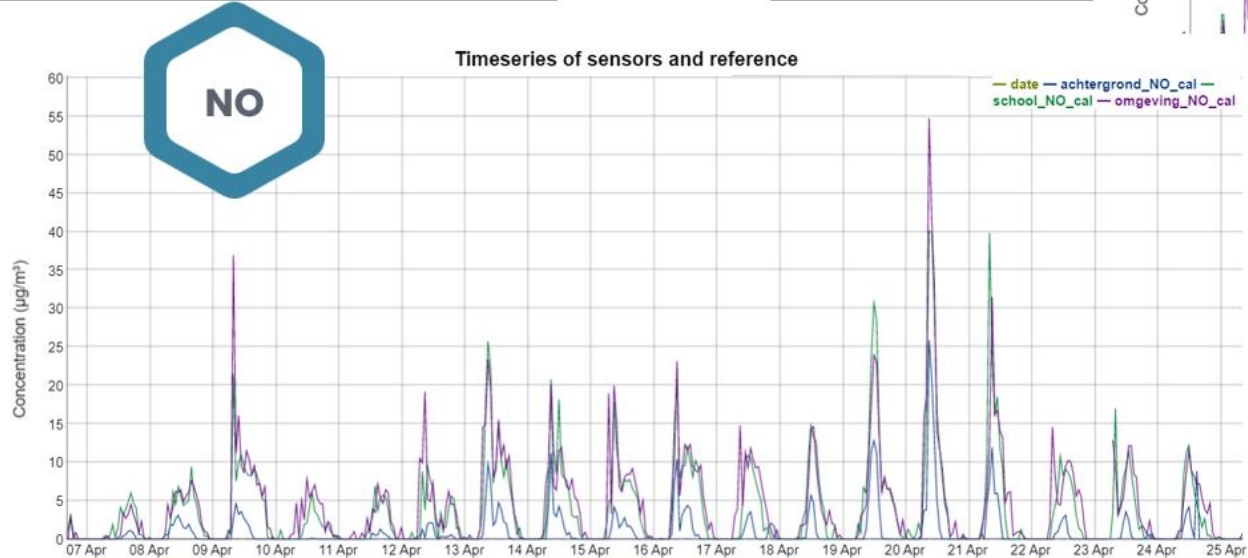
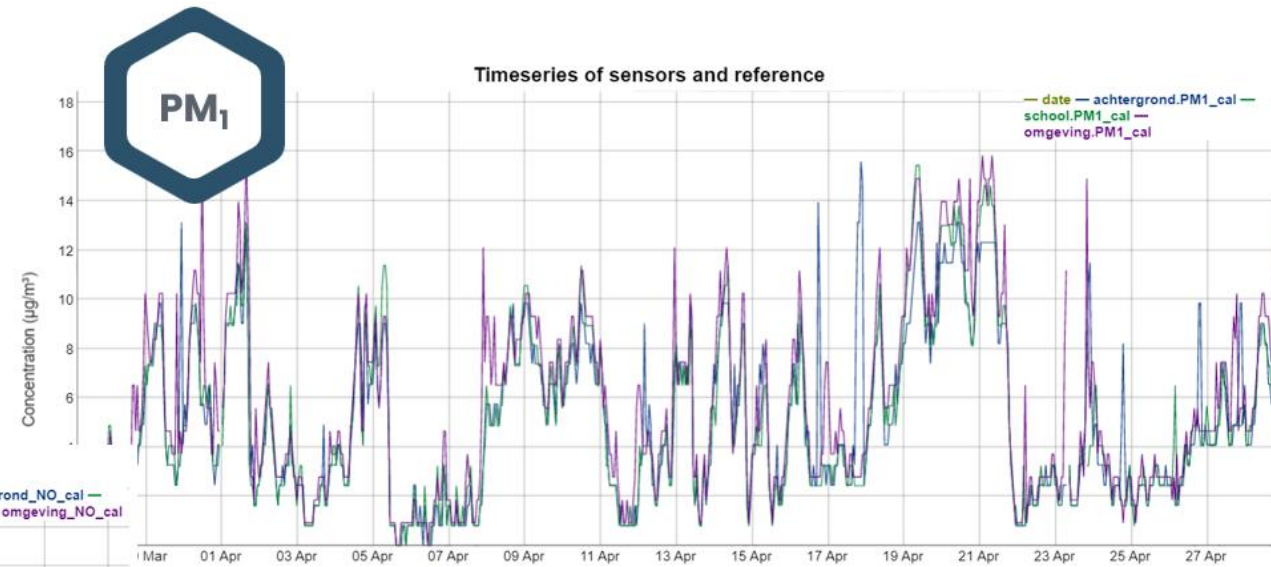
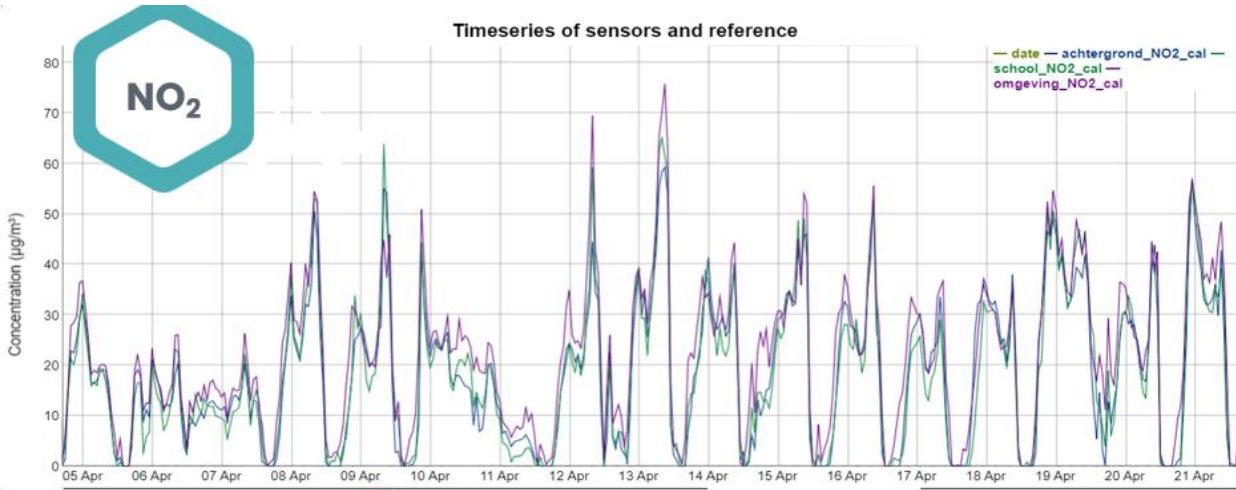


Impact of schoolstreet

- Measurement set-up including:
 - Affected location (study aim! -> traffic measures)
 - Similar unaffected location
 - Background without traffic
- Duration: 13 weeks
- 4 scenarios:
 - School holidays
 - Baseline
 - Reduction 1: knip (1 direction)
 - Reduction 2: schoolstreet

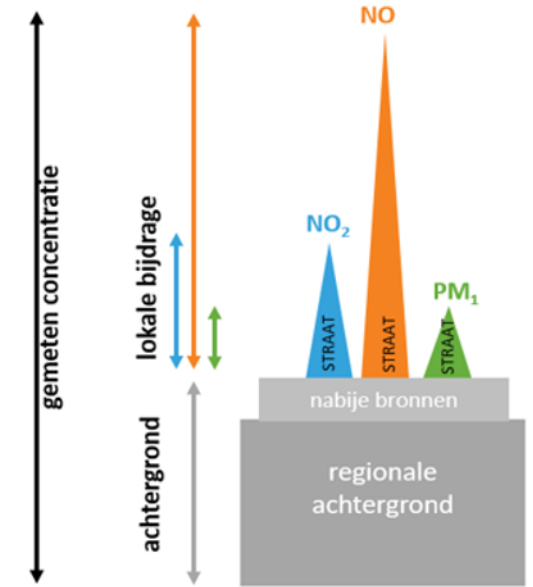
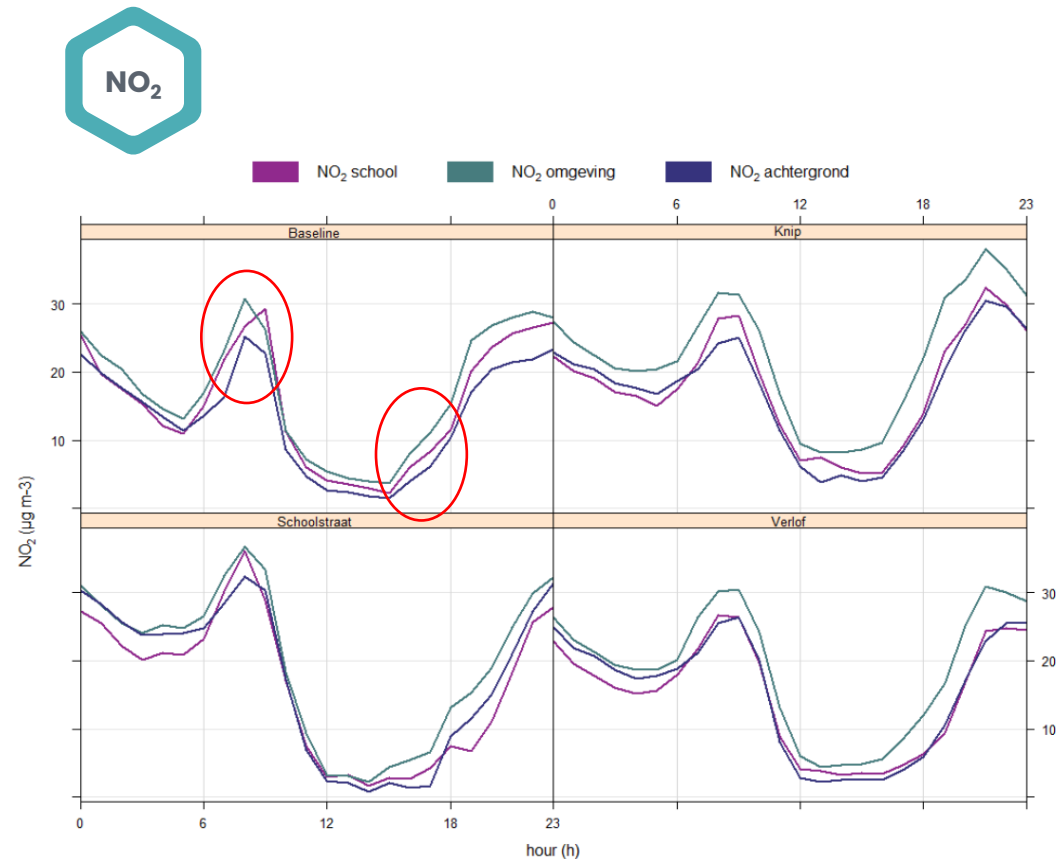


Impact of schoolstreet: time series

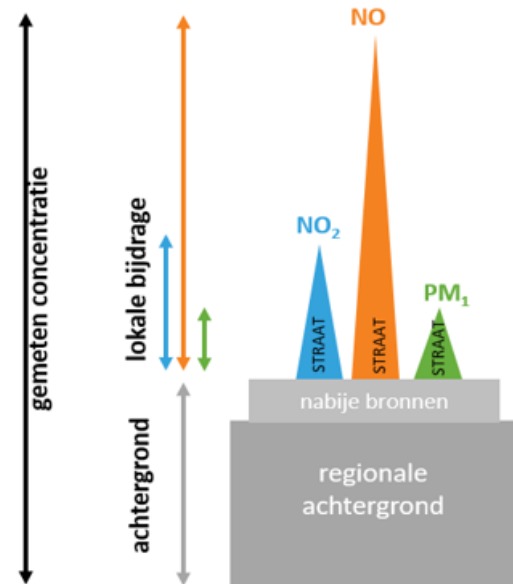
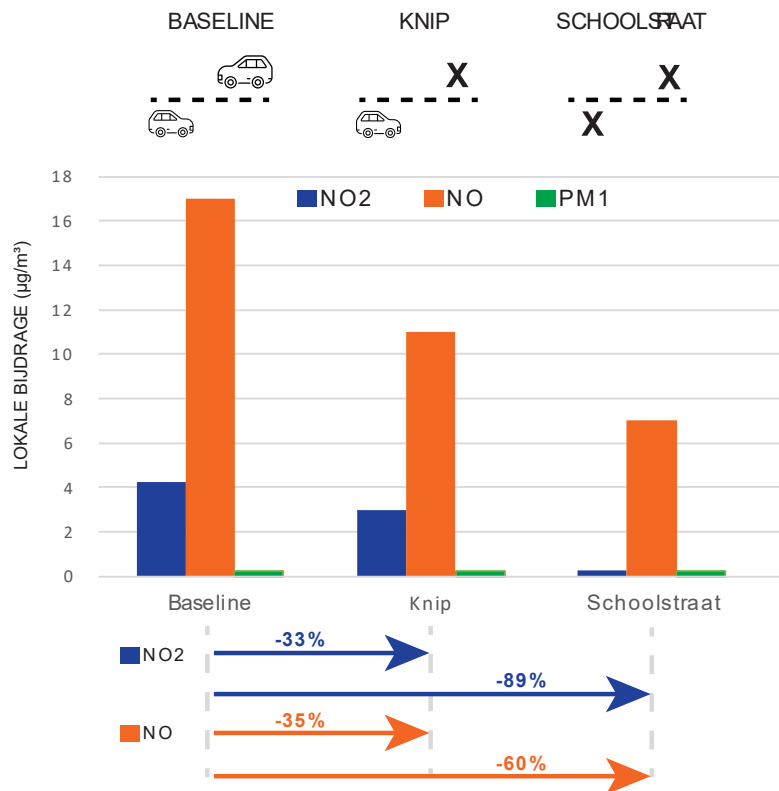
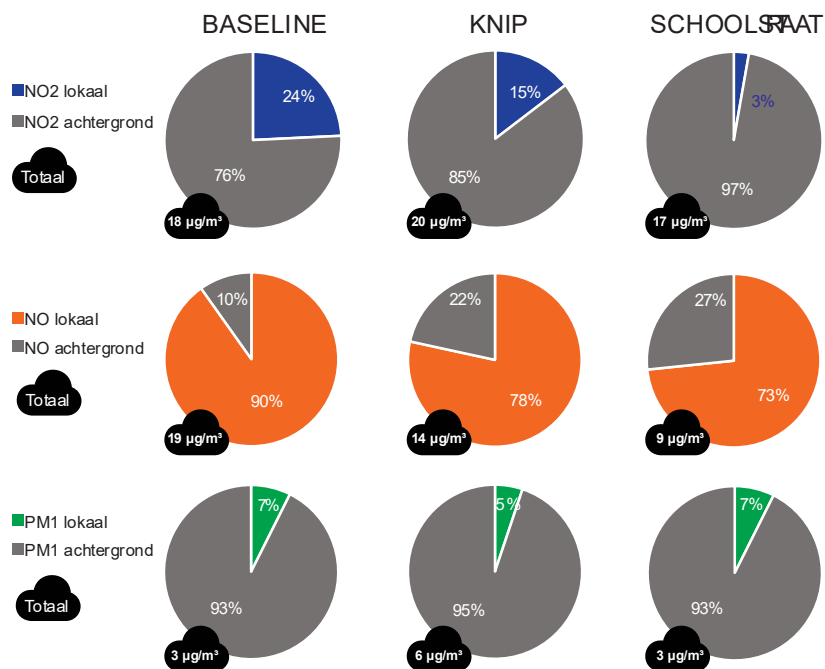


Impact of schoolstreet

- Local contribution!
- Variation of background over time



Impact of schoolstreet



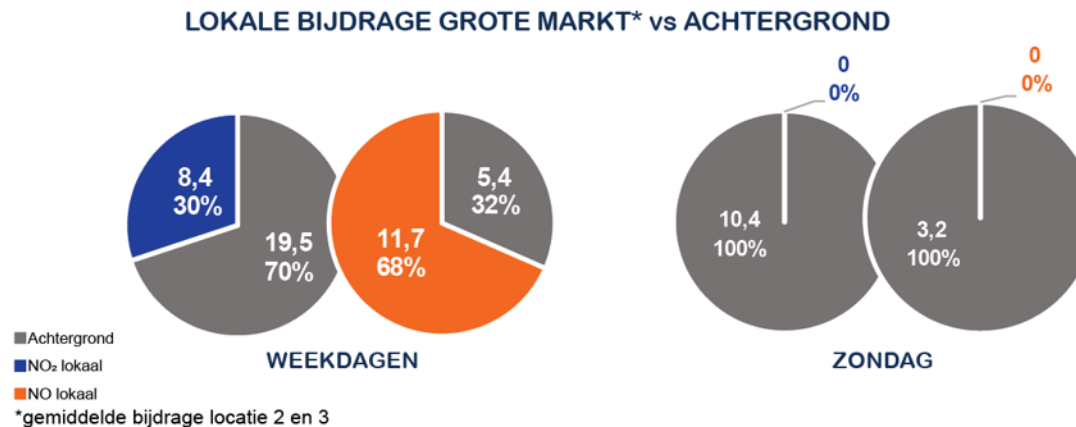
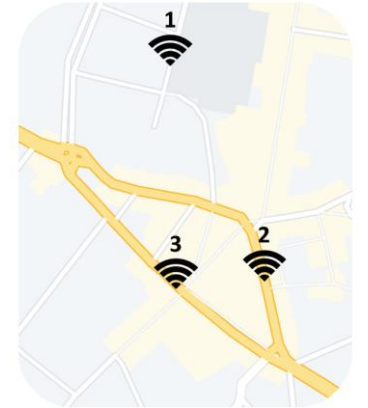
Sint-Niklaas: assessment of traffic impact on major square

- 3 sensor units
- NO, NO₂, PM
- Duration: 2 months
- Include also background concentration!

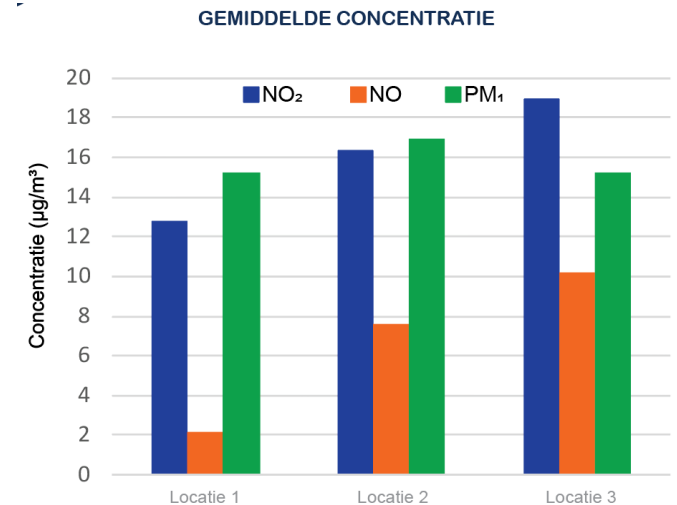


Sint-Niklaas: assessment of traffic impact on major square

- Average NO and NO₂ concentrations are higher at square than at background
- No clear effect for PM₁₀, PM_{2,5}
- Differences between weekdays and weekends



Belangrijke bijdrage van lokaal verkeer aan NO₂ en NO concentraties (PM₁ verwaarloosbaar) ter hoogte van de Grote Markt.
Op zondag zien we geen (of zelfs negatieve) lokale bijdrage.



Data processing levels and use in regulatory monitoring

Data processing levels

Table 1. Proposed Processing Levels for Low-Cost Sensor Systems for Air Quality^a

level	name	definition	example: gas sensors	example: particle sensors
Level-0	raw measurements	original measurand produced by sensor system	voltage corresponding to measured quantity, such as current for electrochemical and infrared sensors, resistance/conductance for metal-oxide sensors	voltage corresponding to current due to light scattered in nephelometers, or to binned counts for optical particle-counters
Level-1	intermediate geophysical quantities	estimate derived from corresponding level-0 data, using basic physical principles or simple calibration equations, and no compensation schemes.	for electrochemical sensors, NO ₂ concentration in µg/m ³ or ppb, using only Level-0 data from the NO ₂ sensor itself with no additional corrections beyond factory calibration ("raw data in concentration units")	binned particle-counts or PM mass in µg/m ³ derived from Level-0 data using simple calibration/assumed particle-density
Level-2A	standard geophysical quantities	estimate using sensor plus other on-board sensors demonstrated as appropriate for artifact correction and directly related to measurement principle ^b	NO ₂ concentration in µg/m ³ or ppb, derived from onboard NO ₂ /NO/O ₃ sensors, corrected for interferences based on the measurement principle and/or T/RH effects using onboard data	PM concentration in µg/m ³ , corrected for T/RH effects with onboard-measured T/RH
Level-2B	standard geophysical quantities-extended	as Level-2A but using external data demonstrated as appropriate for artifact correction and directly related to measurement principle ^b	as Level-2A but using external data from nearby station related to correcting for interferences based on the measurement principle (e.g. O ₃ , T/RH)	as Level-2A but using external T/RH from nearby station
Measurement/Prediction Boundary				
Level-3	advanced geophysical quantities	estimate using sensor plus internal/external inputs, not constrained to data proven as causes of measurement bias or related to measurement principle ^b	NO ₂ concentration in µg/m ³ or ppb, derived from Level-2A or Level-2B data, further corrected by proxies known to be correlated with NO ₂ , e.g. emissions or modeled NO ₂	PM concentration in µg/m ³ , derived from Level-2A or Level-2B data, further corrected by proxies known to be correlated with PM, e.g. emissions or modeled PM
Level-4	spatially continuous geophysical quantities	spatially continuous maps derived from network of sensor systems	map of NO ₂ concentrations in µg/m ³ or ppb, e.g. by assimilation of sensor network data into a physical model	map of PM _{2.5} concentrations in µg/m ³ , e.g. by assimilation of sensor network data into a physical model

Sensor output

Concentrations

Calibration
based on
physical/meas.
principle

Use other proxies to
correct sensor output

Assimilation into a
model

^aT/RH stands for temperature and relative humidity. The spatial support of all Levels except Level-4 is point measurements at single locations or for entire networks. ^bSee Hagler et al. (2018).²

What is an independent sensor measurement?

- Independent sensor measurements: minimum criteria:
 - The concentration output for the target species is mainly determined by the primarily sensor's raw signal;
 - Only corrections that are relevant to the measurement principle (as outlined by ref. Schneider et al.) are allowed;
 - Corrections must also be contemporaneous (i.e., on-line) with the measurement (extending Hagler's point);
 - Data must originate exclusively from the sensor system itself;
 - The calibration model has been developed independently of locally collected environmental data

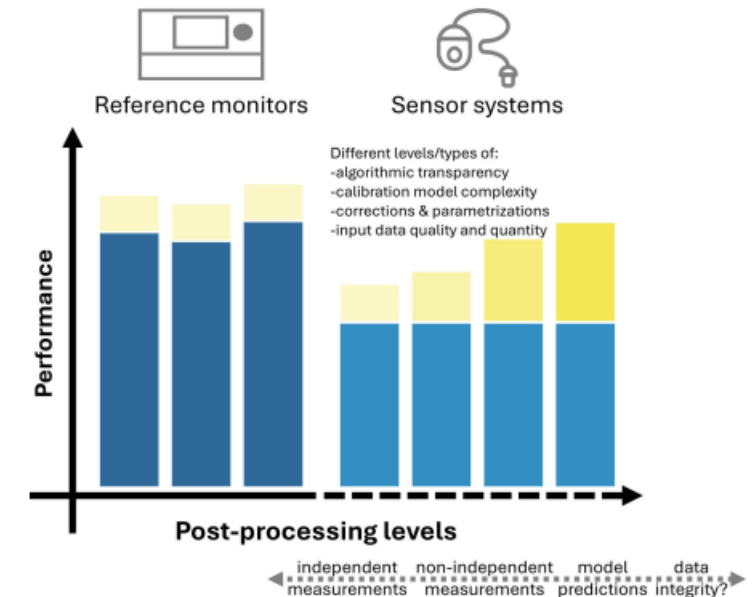


Fig 1 from paper Diez et al., 2025.

EU Air Quality Directive 2008/50/EC

- EU Air Quality Directive: 2008/50/EC
 - Directive on air quality and cleaner air for Europe
 - Objectives for AQ limit values and **common methods** to assess AQ
 - **PM₁₀, PM_{2,5}, NO₂, CO, O₃, SO₂, Pb, Benzene**
 - It includes Data Quality Objectives for measurement techniques:
 - Reference monitoring (fixed sampling points)
 - **Indicative measurements**
 - **Objective estimations**
- **To supplement fixed sampling points**
Sensors can play a role here (but NOT ONLY HERE)



DIRECTIVE (EU) 2024/2881

- “(17) ... The Commission has already mandated work on the preparation of EN standards ... for the evaluation of the performance of sensor systems for the determination of concentrations of gaseous pollutants and particulate matter (PM₁₀ and PM_{2,5}) in ambient air with a view to their **early development and adoption**. In the absence of EN standard methods, the use of ... European Committee for Standardization (CEN) **technical specifications** should be permitted.”

- Indicative measurements
 - data quality objectives that are less strict than those required for fixed measurements
 - to assess AQ in certain zones
 - to provide adequate information on the spatial distribution of air pollutants and on the spatial representativeness of fixed measurements

Some issues with AQ sensors

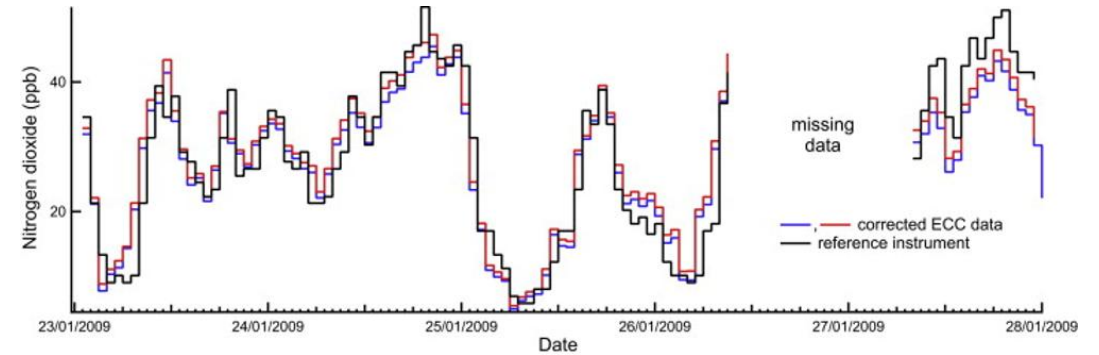
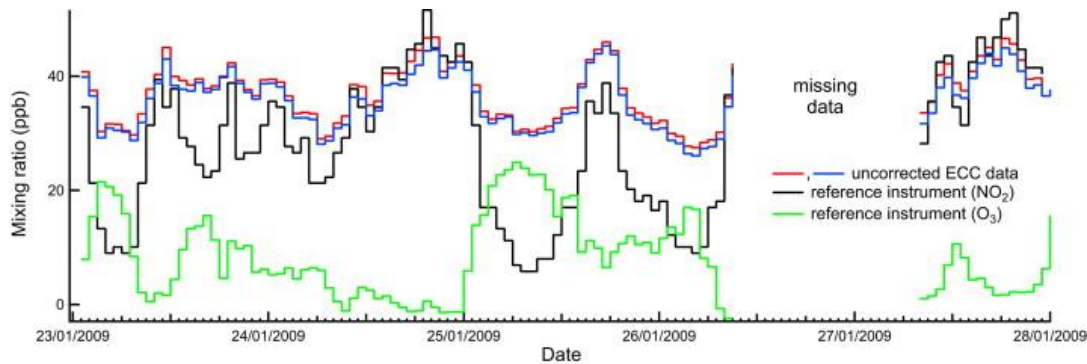
Some issues with AQ sensors

- Selectivity
 - Interference with other pollutants: example of NO₂ and O₃
- PM coarse fraction (PM sensors)
- RH effect
- (T effect)
- Deterioration over time: drift

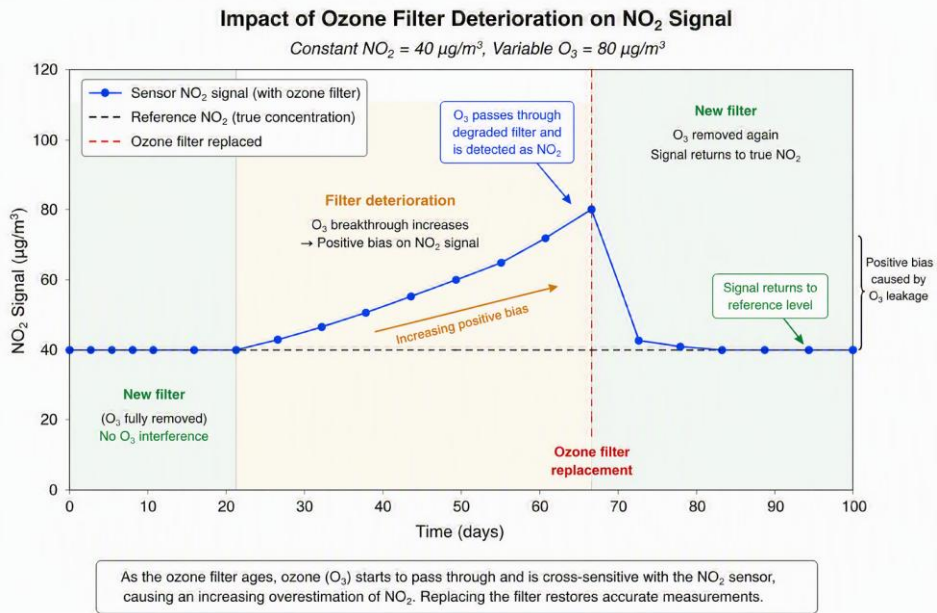
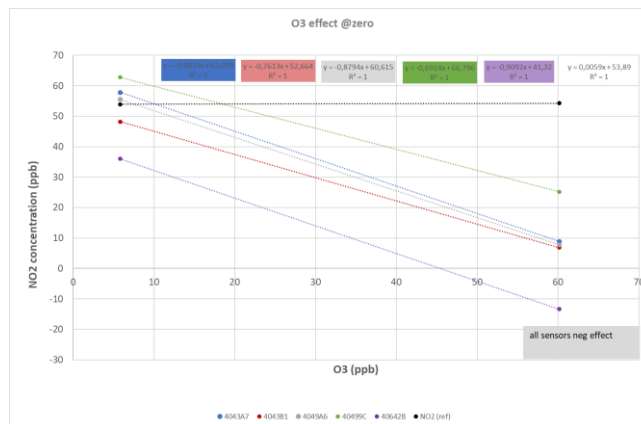
Selectivity: example of NO₂ and O₃

Mead et al. 2013 – correcting NO₂ sensor results for ozone

Interference and correction

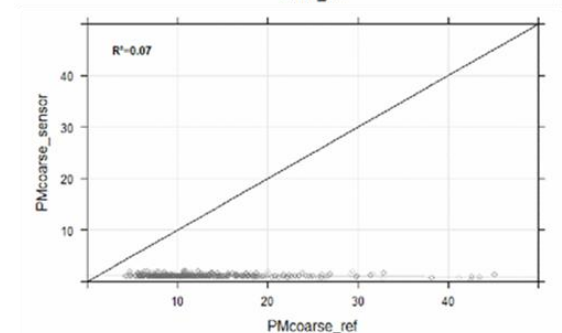
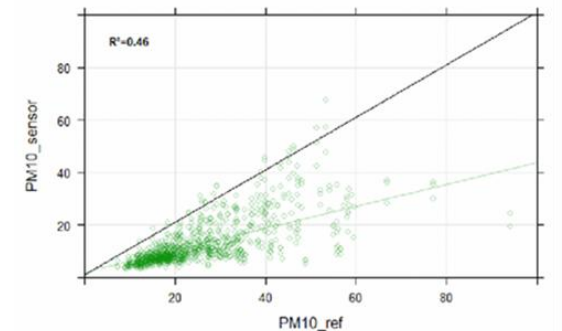
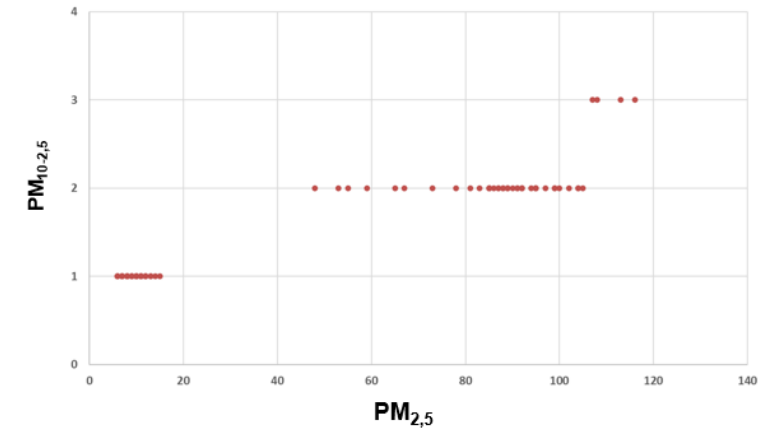


Ozon filter and deterioration



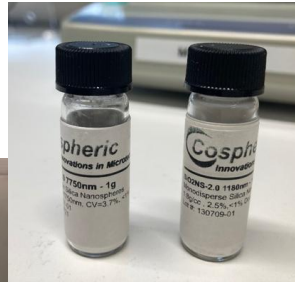
PM coarse problem

- $PM_{coarse} = PM_{10} - PM_{2,5} = PM_{10-2,5}$
- Some sensors do not measure PM_{10} but **calculate PM_{10}** from measured $PM_{2,5}$
- **PROBLEM:**
 - At most locations where sensors are tested, PM_{coarse} is relatively low ($PM_{10} \sim PM_{2,5}$) and PM_{10} values will be acceptable
 - These sensors do **not** pick up PM_{10} peak concentrations at occasions where $PM_{10} \gg \gg \gg PM_{2,5}$
- **SOLUTION:**
 - Simulate high PM_{coarse} conditions in the lab
 - Compare sensor/reference ratio at high and low PM_{coarse} conditions



PM coarse test (lab) - setup

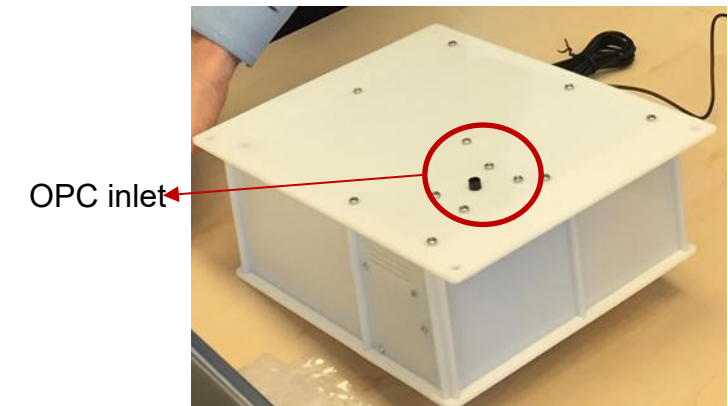
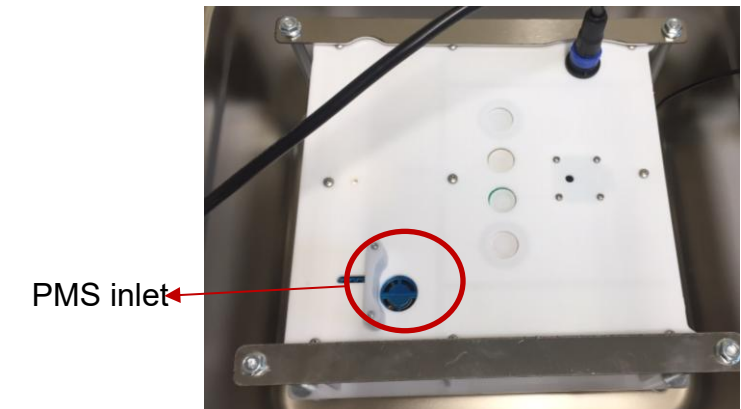
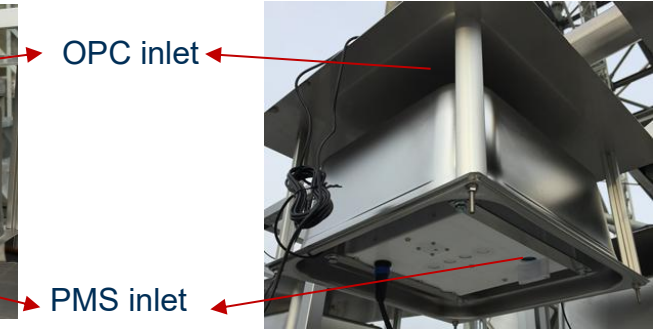
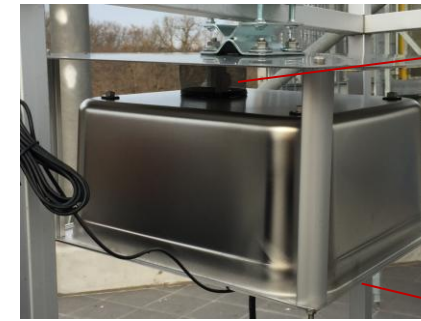
- Two tests with different particle size
 - **Coarse test:** $>80\%$ $PM_{10-2,5}$ in PM_{10}
 - **Fine test:** $< 20\%$ $PM_{10-2,5}$ in PM_{10}
- Sensor has to show a strong response to different fractions, **size restrictions:**
 - Coarse: $5 - 9 \mu m$
 - Fine: $0,4 - 1 \mu m$
- Sensor systems are placed in close vicinity of sampling inlet of the equivalent method (same PM concentration and fractions)
- Change in sensor response from fine to coarse conditions is calculated



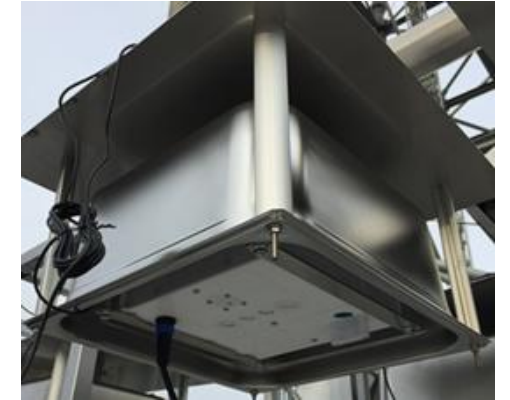
Example: AirsensEUR boxes

- AirSensEUR version 3.0.
- AirSensEUR is an Open Platform project developed by the JRC, aimed at measuring air quality using low-cost sensors
- AirSensEUR includes a PTFE enclosure (26x22x10cm), inserted in a stainless-steel protecting cover (35x32x30cm)
- PM sensors:
 - OPC-N3 with 24 size bins ($0.3/0.35 - 40 \mu\text{m}$) with a counting efficiency* of 50% @ $0.3 \mu\text{m}$ and 100% @ $0.35 \mu\text{m}$
 - PMS5003 with 6 size bins ($> 0.3 \mu\text{m}$) with a counting efficiency* of 50% @ $0.3 \mu\text{m}$ and 100% @ $0.5 \mu\text{m}$

*counting efficiencies claimed by the manufacturers



Example: two different PM sensors



$$R_{PM_{10-2,5}(COARSE)} = \frac{PM_{10-2,5}(sen, COARSE)}{PM_{10-2,5}(eq, COARSE)}$$

$$R_{PM_{2,5}(FINE)} = \frac{PM_{2,5}(sen, FINE)}{PM_{2,5}(eq, FINE)}$$

$$F\left(\frac{FINE}{COARSE}\right) = \frac{R_{PM_{2,5}(FINE)}}{R_{PM_{10-2,5}(COARSE)}}$$

< 2 for Class 1!

<2 →

0.27	0.92
0.06	1.42
4.3	0.65

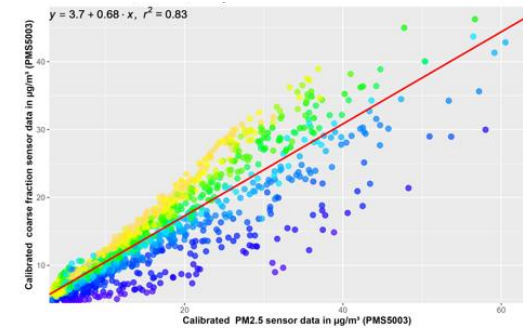
>2
FAIL

<2
PASS

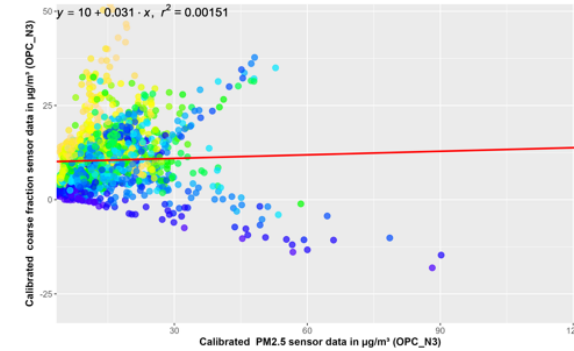
AirsensEUR boxes: field data

- $PM_{10-2,5}$ versus $PM_{2,5}$
 - No linear relationship observed for equivalent (reference): $PM_{10-2,5}$ versus $PM_{2,5}$
 - PMS5003 shows linear relationship
- PMS sensor versus equivalent (reference):
 - Clear linear relationship for $PM_{2,5}$ no linear relationship for $PM_{10-2,5}$
- Observations of lab tests are confirmed

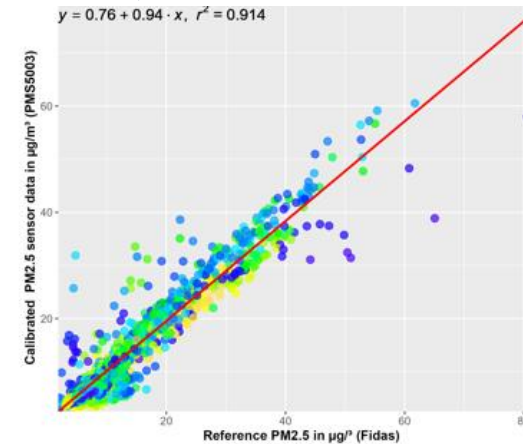
$PM_{10-2,5}$ vs $PM_{2,5}$: PMS



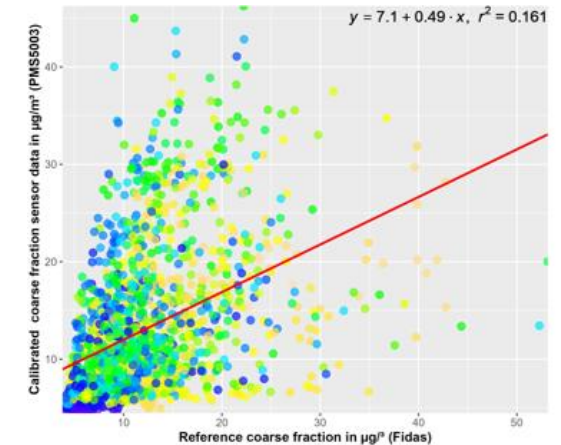
$PM_{10-2,5}$ vs $PM_{2,5}$: OPC



$PM_{2,5}$: PMS vs reference

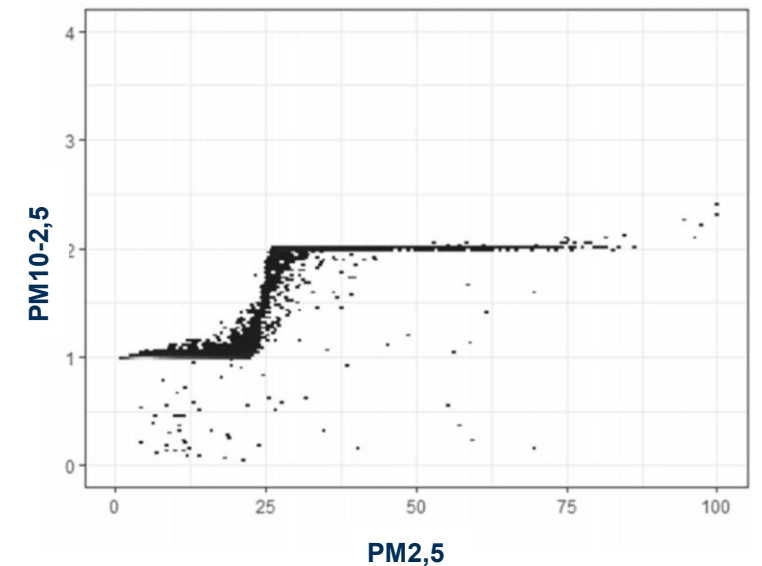


$PM_{10-2,5}$: PMS vs reference



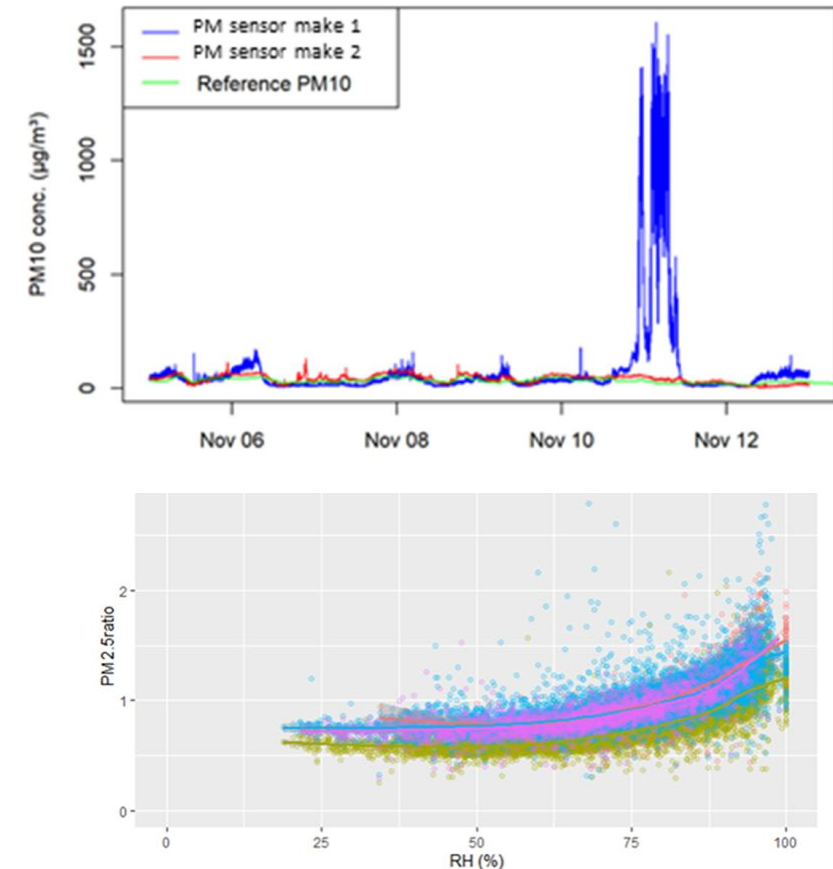
PM_{coarse} plotting field data: what can you learn?

- Plot PM_{10-2,5} of sensor system against the reference method.
- Plot sensor values PM_{10-2,5} versus PM_{2,5}
 - Use hourly data to see if there are any trends
- Evaluation:
 - PM_{10-2,5} (sensor vs reference): $0,5 < \text{slope} < 2$; $R^2 > 0,5$
 - No pattern identified



RH problem

- Some sensors overestimate PM readings at high RH
- **PROBLEM:**
 - When only looking at 24h values 'over' readings can be compensated by 'under' readings during other parts of the day
 - No RH issue will be observed based on 24h averages whereas strong overestimations can occur at hourly values
- **SOLUTION:**
 - Plot hourly average sensor readings as function of RH
 - Compare sensor/reference ratio at 'dry' and 'humid' conditions in the field



Examples from VAQUUMs and sensEURcity projects

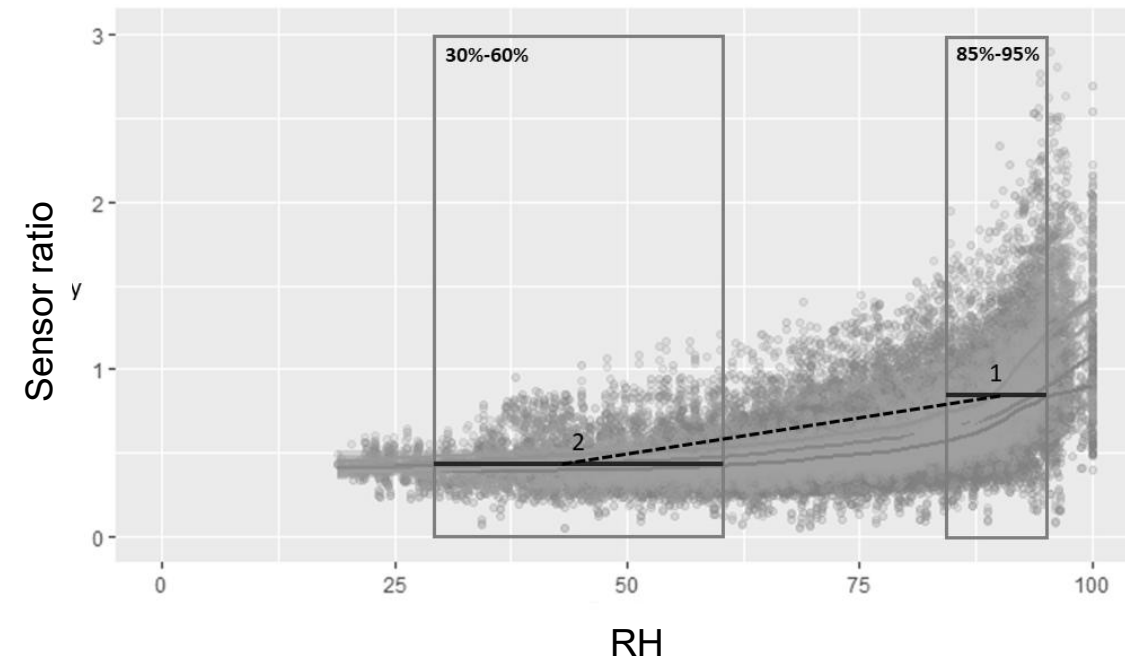
RH effect (calculations)

- Field data
- Calculate sensor ratio
 - ratio of sensor value to equivalent value
 - using hourly values
- RH factor is defined as change in sensor ratio from low (30-60%) to high (85-95%) RH

$$RH_{factor} = \frac{Mdn(sensorratio)_{85-95\%}}{Mdn(sensorratio)_{30-60\%}}$$

- Evaluation for each sensor system and test site
 - at least 20 hourly values in each RH range

$$Sensor\ ratio = \frac{y_{sen}}{y_{eq}}$$



$RH_{factor} < 1.5$: OK

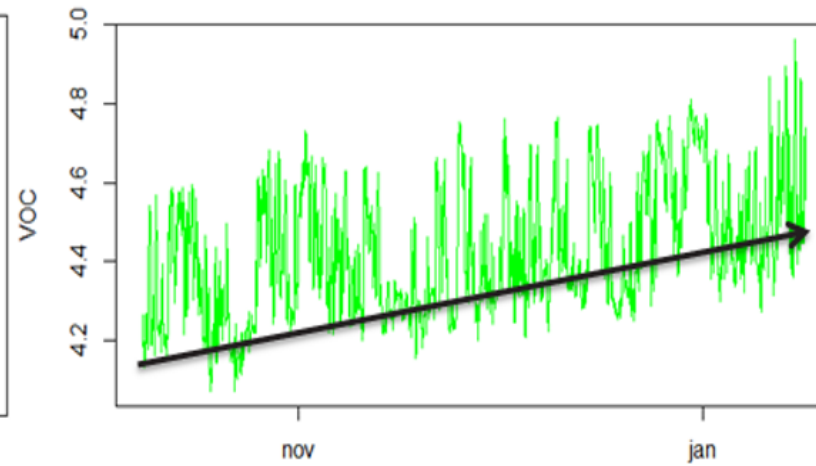
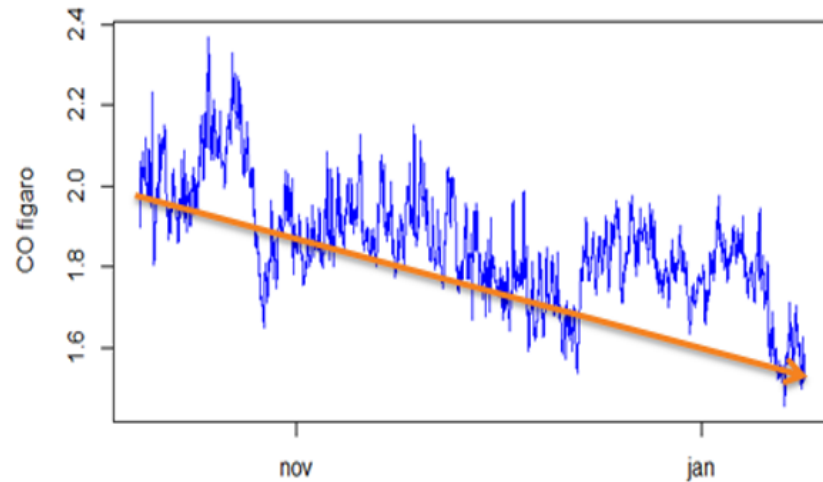
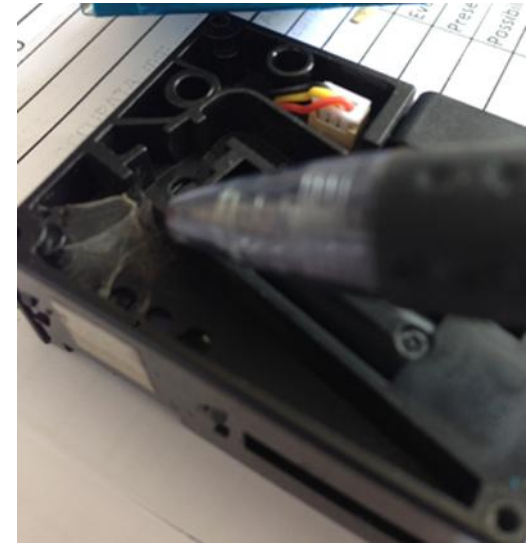
$1.5 < RH_{factor} < 2$: less performing sensor

RH effect plotting field data: what can you learn?

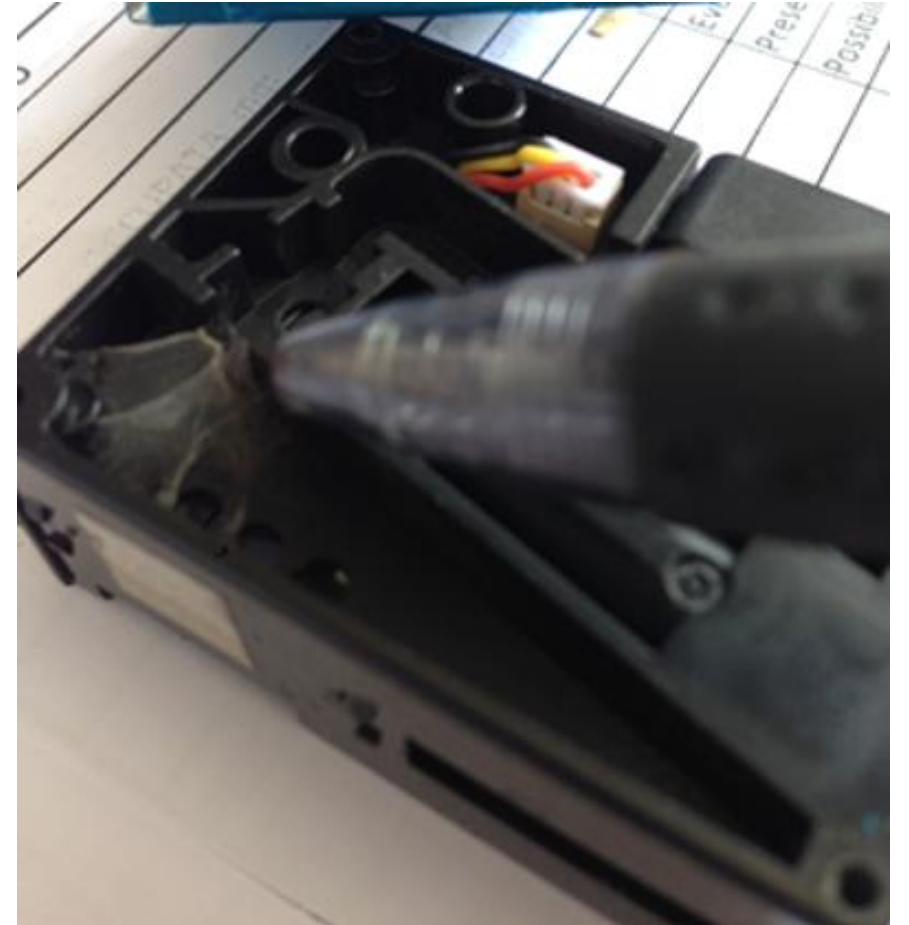
- Plot PM of sensor system against the RH
 - Calculations (see previous slide)
 - Compare with trend of equivalent method
- Plot PM of sensor system vs equivalent (ref) method (hourly values)
 - Optional with color-code ifo RH to see trends (overestimating/underestimating)

Sensor drift

- Deterioration of sensors over time resulting in drift
- Frequent checks are needed: eg by co-location, by checking background signals of sensors , etc...
- Solutions: Sensor replacement, recalibration, cleaning (eg PM sensors)

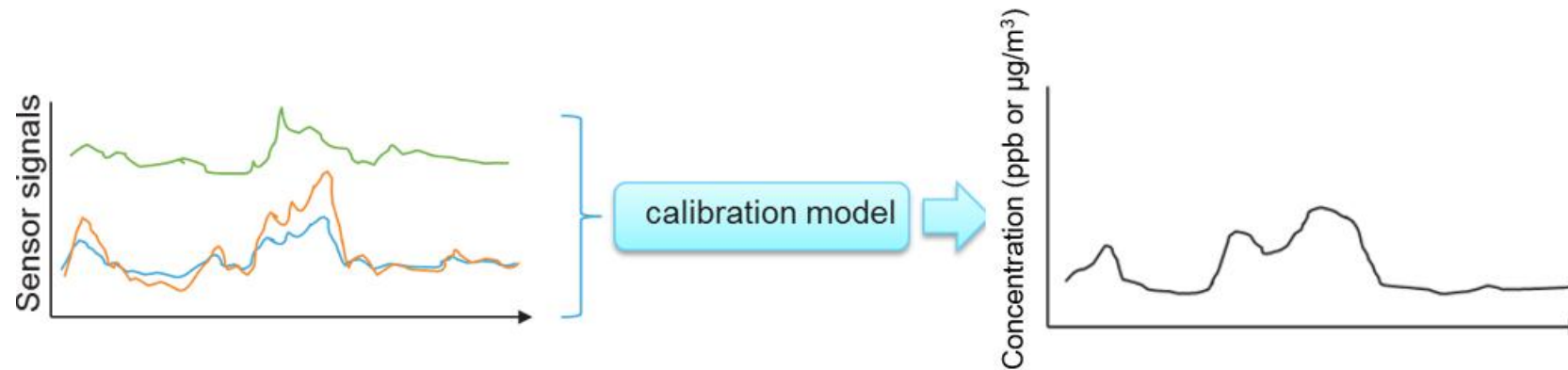
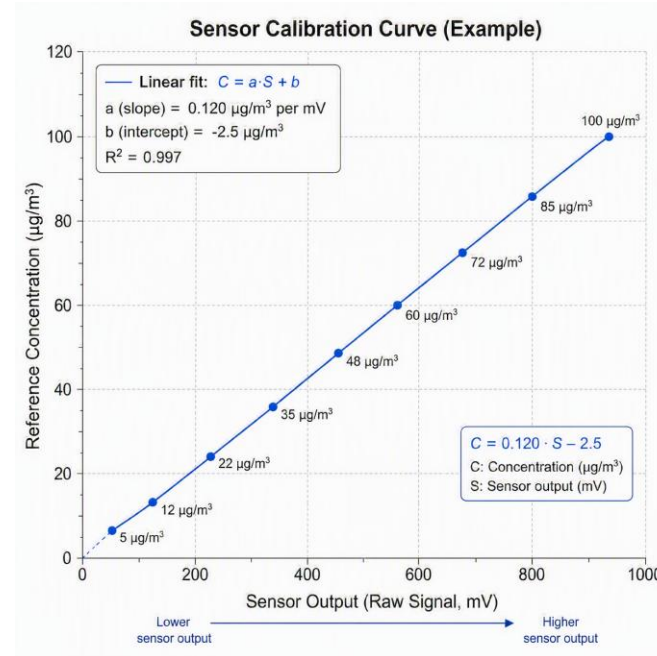


Outliers



Sensor calibration

Calibration: what?



Definitions

- Calibration:
 - Define/set the calibration model parameters to adjust the sensor response based on relation between reference concentration and sensor measured concentration
- Correction:
 - Adjusting sensor response to compensate for systematic effects e.g. for environmental influences (this can be done without reference)
 - For PM sensors, correction can be applied to adjust for local aerosol properties and humidity effects
 - Correction can also be applied to align readings of different sensors (without reference), this is often referred to as 'rescaling'
- Verification/validation:
 - Control the sensor concentrations to confirm calibration
 - When training calibration models there is always a training period and a verification

Difference with reference analysers

- Reference analysers:
 - Calibration is checked on a regular basis (e.g. daily)
 - Connect a bottle with known concentration to monitor inlet and adapt reading
- Sensors:
 - Typically have **no (active sampling) inlet** (even if build in a sensor system)
 - Need to be exposed to the target concentration in test chamber



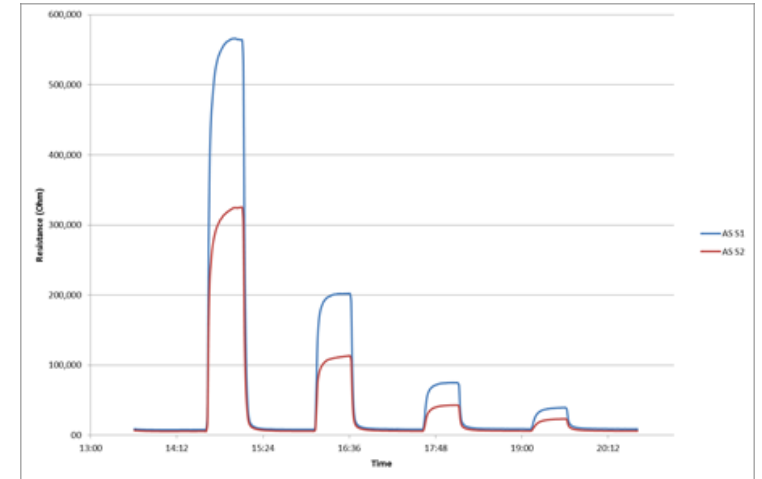
Laboratory versus field calibration

- Lab calibration:
 - Controlled: concentration range,...
 - Known interferences, meteorological conditions,...
 - Not necessary representative conditions
- Field calibration
 - Real conditions
 - Unknown interferences are taken into account but no insight in effect
 - Transfer other locations?



Calibration/correction models (see also data processing)

- Different models
 - Linear model
 - Simple linear regression (only function of concentration)
 - Multiple linear regression with environmental variables
 - Non-linear model
 - Machine learning (ML) models
- Parameters
 - sensor output
 - Temperature
 - relative humidity
 - Pressure
 - Other pollutant concentrations (interferents)
 - Other variables



What about ‘calibration’ of sensor networks?

- Sensor networks can use different calibration methods
 - Co-location of sensor systems with reference analysers at selected locations can be used to ‘calibrate’ the wider network
 - Remote calibration using background correction
 - Remote calibration at periods with similar concentrations
 - ...
- Some examples will be given

